

Optimizing Latency in Cloud Gaming Through Edge Computing and Micro Data Centers: A Simulation-Based Study

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I. ABSTRACT

The rise of online multiplayer video games is something we have seen drastically increasing at an expanding rate. In comparison to 2018 when online multiplayer games only consisted of PC games such as Fortnite, H1Z1 and PUBG (Player Unknown Battlegrounds) to now the customer base which includes mobile, PC and console games. In 2025, it was recorded that the customer base consists of more than 500 million live players. But with this rise in customer base the technology has not kept up especially in a country like India. The problem statement that this paper states is that the latency that a player receives while playing video games is too high and can be improved. This paper focuses on the scenario where a player is sitting in any one of the four-metropolitan cities and playing a video game whose server is in Singapore. This paper tries to solve the problem of high latency faced by the players through the approach of installing Modular Edge Data Centers (MDCs). This framework aims to improve the Quality of Experience (QoE) for this rapidly expanding user base.

II. INTRODUCTION

A. Growth of Cloud Gaming in India

India has seen an improved growth in online gaming led by increased technological advancement and greater internet penetration due to an increase in player base. With over 500 million gamers, our country is one of the largest gaming markets. This rise has resulted in an increase in need of more Cloud-Streamed Gaming Services, which allows the user to access high quality games without needing the required hardware. As per the past data, market research shows that this number is expected to grow even more in the coming decades, supported by nationwide 5G service deployment. Internet Service Providers (ISPs) and Telecom companies are constantly investing in infrastructure to meet the increasing demand and expanding user base and ensure network reliability and Quality of Service (QoS).

B. The Latency Challenge in Existing Architectures

Despite the rapid digital expansion, network latency remains a critical problem to delivering high quality service. Latency, also measured as Round-Trip Time (RTT) directly affects directly impacts gameplay performance especially in First Person Shooter (FPS) games and esports titles. High latency or RTT results in delayed responses, poor synchronization, and reduced Quality of Service (QoS). This problem was faced by me and many of my peers in our childhood when we were involved in playing multiplayer games. Now that I have a more profound understanding of Computer Networks and experience in the field of Computer Science I wanted to research on this problem and try to study upon a solution that could improve user experience and for other people not to face the same problem that I did. The primary reason for facing such issues is geographical distance. Due to India's demographic video game companies do not prefer to expand their video game servers to a country like India leading to server establishment overseas in countries like Singapore for servers in Southeast Asia. Moreover, servers set up locally inefficient ISP routing, and poor peer to peer connections result in delays even when the bandwidth is high.

C. Research Contributions

This paper contains a comprehensive study on implementing Ultra Low Latency (ULL) through a structured and simulation using Cisco Packet Tracer. The simulation contains two scenarios, scenario 1 takes the example of 4 gamers sitting in separate cities in India trying to connect to a server in Singapore with a latency of around 100-150 ms whereas in Scenario 2 we try to use the approach of using Modular Edge Data Centers within the country to improve latency and better routing mechanisms. I have used Cisco Packet Tracer to simulate the 2 scenarios.

III. LITERATURE REVIEW

A. Impact of Latency on Quality of Experience

Latency, also known as Round Trip Time (RTT) is one of the most important factors to determine Quality of Experience for the user in online multiplayer gaming. Users have reported

that the Round-Trip Time (RTT) anywhere around 30-60 ms is playable whereas RTT more than 100 ms really degrades user gameplay performance resulting in slower response time and increased frustration among the players. Especially for competitive games including FPS (First Person Shooter) games, this problem could be devastating for the user. For such games, the RTT should be below 60 ms. Other than RTT, there are several other factors that the gamers depend on, which is network stability which is affected by Packet Loss and Network Jitters. Constant packet loss and network jitters also affect user sessions. Hence, they are also very crucial to stabilize other than RTT.

B. Server Proximity and Geographic Determinism

Physical and network proximity are a very crucial factor that should affect latency. Prior studies have shown that RTT increases proportionally with geographic distance that affects fairness, responsiveness, and retention. In the context of India ISP companies such as Jio and Airtel partnering up with Local ISPs have been able to penetrate the population, but deployment of local data centers is still lacking, as a result user face difficulty connecting to international servers especially South-East Asia leading to high latency and inconsistent performance.

C. Infrastructure and Policy Constraints

Beyond geographical factors, infrastructural and policy related factors impact Quality of Service (QoS). Even when servers are locally located, many unavoidable delays still happen due to inefficient ISP routing and weak peering agreements. Economic constraints result in decentralized deployment. High power tariffs and increase in Total Cost of Ownership (TCO) cause increased infrastructural clustering in Tier-1 metropolitan cities as compared to Tier-2 and Tier-3 cities.

IV. METHODOLOGY

A. Research Design and Experimental Framework

This research uses Cisco Packet Tracer to simulate cloud gaming network architectures.

Two scenarios were implemented:

- Centralized Cloud Architecture
- Edge Computing Architecture with MDC

Cisco Packet Tracer was chosen as the simulation environment because of its support for dynamic routing protocols, multi-router backbone emulation, and packet-level traffic visualization.

B. Theoretical Basis for Network Architecture Design

The network topology is modeled after a hierarchical architecture commonly used in real world network simulation scenarios. The layers are:

- Access Layer (Home WiFi Routers)
- Regional ISP Layer
- National Core Layer
- International Transit Layer

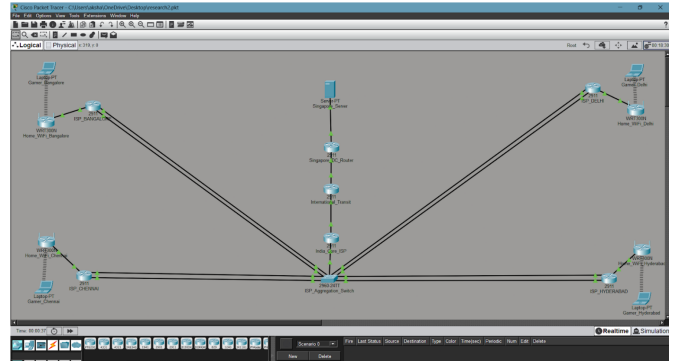


Fig. 1. Scenario 1 Network Topology

• Data Center Layer

The principal behind these layers is that the network latency increases with each layer due to an increase in number of intermediate hops. The topology for Scenario 1 consists of 4 home laptops connected to their home Wi-Fi networks which are connected to a regional ISP which is then connected to a national core ISP to an International Transit Router and finally the Singapore Data Center. Now scenario 2 contains the same topology although there is one very major difference in the second scenario there is a Modular Edge Data Center is put directly connected to the India Core ISP. I have provided the pictures for the same.

C. Scenario 1: Centralized Cloud Model Implementation

Cloud based games allow users to play high quality games without the need to have high performance hardware by streaming the game directly from a remote cloud-based server. Moreover, greater geographical distance causes network delays and latency problems. This scenario simulates a traditional cloud-based setup where 4 gamers from 4 separate Indian cities connect to a centralized cloud gaming server in Singapore. Cisco Packet Tracer was used to simulate this scenario.

The objectives of this simulation are:

- To design a multi-city cloud gaming network
- To simulate real-world ISP and backbone routing
- To connect gamers to an international cloud server
- To analyze packet flow and latency
- To establish a base model for optimization comparison

1) Devices Used: End Devices

- 4 Gamer Laptops
- Singapore Cloud Gaming Server

Network Devices

- 4 Home WiFi Routers (WRT300N)
- 4 ISP Routers (Cisco 2911)
- India Core ISP Router
- International Transit Router
- Singapore DC Router
- Aggregation Switch (Cisco 2960)

```

Router>
Router>enable
Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address        Interface
20.0.0.1        1    2WAY/DROTHER    00:00:34    10.0.0.254     GigabitEthernet0/2
192.168.30.1    1    FULL/BDR        00:00:34    10.0.0.3        GigabitEthernet0/2
192.168.20.1    1    2WAY/DROTHER    00:00:35    10.0.0.2        GigabitEthernet0/2
192.168.40.1    1    FULL/DR         00:00:34    10.0.0.4        GigabitEthernet0/2
Router#

```

Fig. 2. OSPF Neighbor Table on ISP_Bangalore

```

Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address        Interface
192.168.10.1    1    2WAY/DROTHER    00:00:30    10.0.0.1        GigabitEthernet0/2
20.0.0.1        1    2WAY/DROTHER    00:00:29    10.0.0.254     GigabitEthernet0/2
192.168.30.1    1    FULL/BDR        00:00:30    10.0.0.3        GigabitEthernet0/2
192.168.40.1    1    FULL/DR         00:00:30    10.0.0.4        GigabitEthernet0/2
Router#

```

Fig. 3. OSPF Neighbor Table on ISP_Delhi

```

Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address        Interface
20.0.0.1        1    FULL/DROTHER    00:00:35    10.0.0.254     GigabitEthernet0/2
192.168.20.1    1    FULL/DROTHER    00:00:36    10.0.0.2        GigabitEthernet0/2
192.168.40.1    1    FULL/DR         00:00:35    10.0.0.4        GigabitEthernet0/2
192.168.10.1    1    FULL/DROTHER    00:00:35    10.0.0.1        GigabitEthernet0/2
Router#

```

Fig. 4. OSPF Neighbor Table on ISP_Hyderabad

```

Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address        Interface
192.168.10.1    1    FULL/DROTHER    00:00:31    10.0.0.1        GigabitEthernet0/2
20.0.0.1        1    FULL/DROTHER    00:00:31    10.0.0.254     GigabitEthernet0/2
192.168.30.1    1    FULL/BDR        00:00:31    10.0.0.3        GigabitEthernet0/2
192.168.20.1    1    FULL/DROTHER    00:00:31    10.0.0.2        GigabitEthernet0/2
Router#

```

Fig. 5. OSPF Neighbor Table on ISP_Chennai

```

Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address        Interface
30.0.0.2        1    FULL/DR         00:00:35    20.0.0.2        GigabitEthernet0/1
192.168.40.1    1    FULL/DR         00:00:35    10.0.0.4        GigabitEthernet0/0
192.168.10.1    1    2WAY/DROTHER    00:00:35    10.0.0.1        GigabitEthernet0/0
192.168.30.1    1    FULL/BDR        00:00:35    10.0.0.3        GigabitEthernet0/0
192.168.20.1    1    2WAY/DROTHER    00:00:36    10.0.0.2        GigabitEthernet0/0
Router#

```

Fig. 6. OSPF Neighbor Table on ISP_India_Core

```

Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address        Interface
172.16.1.1      1    FULL/DR         00:00:38    30.0.0.1        GigabitEthernet0/1
20.0.0.1        1    FULL/BDR        00:00:38    20.0.0.1        GigabitEthernet0/0
Router#

```

Fig. 7. OSPF Neighbor Table on International Transit Router

```

Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address        Interface
30.0.0.2        1    FULL/BDR        00:00:36    30.0.0.2        GigabitEthernet0/0
Router#

```

Fig. 8. OSPF Neighbor Table on Singapore DC Router

2) IP Addressing Scheme:

TABLE I
IP ADDRESSING SCHEME

Device	Network	Gateway
Gamer Bangalore	192.168.10.0/24	192.168.10.1
Gamer Delhi	192.168.20.0/24	192.168.20.1
Gamer Hyderabad	192.168.30.0/24	192.168.30.1
Gamer Chennai	192.168.40.0/24	192.168.40.1
Singapore Server	172.16.1.100/24	172.16.1.1
ISP Bangalore	10.0.0.1/24	-
ISP Delhi	10.0.0.2/24	-
ISP Hyderabad	10.0.0.3/24	-
ISP Chennai	10.0.0.4/24	-
ISP India Core	20.0.0.1/24	-
International Transit	20.0.0.2/24	-
Singapore DC Router	30.0.0.1/24	-

3) *Routing Configuration:* OSPF dynamic routing protocol was configured on:

- City ISP Routers
- India Core Router
- International Transit Router
- Singapore DC Router

This allowed automatic route propagation across the network.

OSPF Neighbor Tables

1. ISP_Bangalore
2. ISP_Delhi
3. ISP_Hyderabad
4. ISP_Chennai
5. ISP_India_Core
6. International Transit
7. Singapore_DC_Router

4) *Limitations of Traditional Architecture:* The limitations of traditional centralized cloud gaming architecture are:

- High latency due to long-distance routing
- Multiple ISP hops
- Dependence on international servers
- Increased delay for real-time gaming
- Potential packet loss and jitter

D. Scenario 2: Edge Computing and MDC Integration

As we saw in Scenario 1, we simulated a Traditional Architecture completely depend on centralized servers located at a distance from the user which results in high network latency (RTT) and delays leading to a degraded user performance. In this Scenario 2, we use the architecture of Modular Edge Data Centers (MDC) and Optimized Routing to overcome the limitations we experience in Scenario 1.

The proposed architecture consists of:

- A local Edge Data Center within Indian Region
- A Modular Edge Data Center (MDC) router connected to the India_Core_ISP
- An Edge Game Server for local processing
- Optimized routing to prioritize local traffic

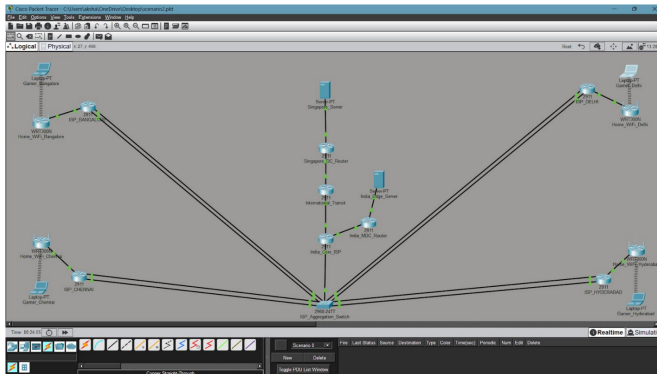


Fig. 9. Scenario 2 Network Topology

An Edge Data Center is added within the Indian region to simulate a local gaming network. A dedicated Modular Edge Data Center is added between the Core ISP Indian router and the International Transit. This ensures that the local networking requests are processed locally without travelling to international servers.

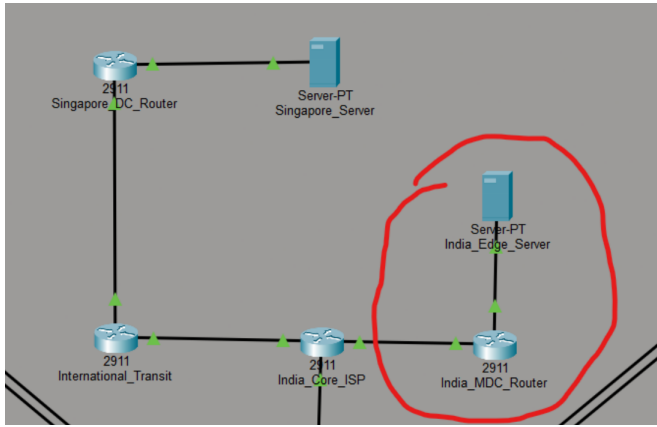


Fig. 10. Scenario 1 Network Topology

1) *IP Addressing and Configuration:* The Edge network was configured with a dedicated subnet:

- Edge Network: 192.168.200.0/24
- Edge Server IP: 192.168.200.10
- MDC Gateway: 192.168.200.1

A point-to-point connection between India Core and MDC router was configured:

- Core-MDC Link: 40.0.0.0/30

2) *Optimized Routing Implementation:*

- Lower OSPF cost assigned to MDC/Edge path
- Higher cost assigned to international cloud path
- Ensured gaming traffic prefers local edge server

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.40.1	1	FULL/DR	00:00:30	10.0.0.4	GigabitEthernet0/0
192.168.10.1	1	2WAY/DROTHER	00:00:30	10.0.0.1	GigabitEthernet0/0
192.168.30.1	1	FULL/DR	00:00:30	10.0.0.3	GigabitEthernet0/0
192.168.20.1	1	2WAY/DROTHER	00:00:30	10.0.0.2	GigabitEthernet0/0
192.168.200.1	1	FULL/DR	00:00:37	40.0.0.2	GigabitEthernet0/2
30.0.0.1	1	FULL/DR	00:00:30	20.0.0.2	GigabitEthernet0/1

Router#

Fig. 11. OSPF Routing of MDC on India_Core_ISP

3) *Impact of Proposed Solution:* The proposed architecture demonstrates how integrating:

- Edge computing
- Micro data centers
- Optimized routing
- 5G-enabled infrastructure

V. RESULTS

A. Scenario 1: Traditional Cloud Architecture

1) *Connectivity Testing:* After configuring the connectivity was verified by two methods, one was by pinging the Singapore Server and the other was by simulating movement of a packet from the Gamer Laptops to:

- Local Wi-Fi router
- Local ISP router
- India Core router
- Singapore cloud server

<pre>C:\>ping 10.0.0.1</pre> Pinging 10.0.0.1 with 32 bytes of data: <pre>Reply from 10.0.0.1: bytes=32 time=25ms TTL=254 Reply from 10.0.0.1: bytes=32 time=12ms TTL=254 Reply from 10.0.0.1: bytes=32 time=20ms TTL=254 Reply from 10.0.0.1: bytes=32 time=5ms TTL=254</pre> Ping statistics for 10.0.0.1: <pre>Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 5ms, Maximum = 25ms, Average = 15ms</pre>	<pre>C:\>ping 10.0.0.2</pre> Pinging 10.0.0.2 with 32 bytes of data: <pre>Reply from 10.0.0.2: bytes=32 time=30ms TTL=254 Reply from 10.0.0.2: bytes=32 time=20ms TTL=254 Reply from 10.0.0.2: bytes=32 time=20ms TTL=254 Reply from 10.0.0.2: bytes=32 time=5ms TTL=254</pre> Ping statistics for 10.0.0.2: <pre>Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 5ms, Maximum = 30ms, Average = 19ms</pre>
Gamer_Bangalore to ISP_Bangalore <pre>C:\>ping 10.0.0.3</pre> Pinging 10.0.0.3 with 32 bytes of data: <pre>Reply from 10.0.0.3: bytes=32 time=15ms TTL=254 Reply from 10.0.0.3: bytes=32 time=15ms TTL=254 Reply from 10.0.0.3: bytes=32 time=25ms TTL=254 Reply from 10.0.0.3: bytes=32 time=13ms TTL=254</pre> Ping statistics for 10.0.0.3: <pre>Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 13ms, Maximum = 25ms, Average = 16ms</pre>	Gamer_Delhi to ISP_Delhi <pre>C:\>ping 10.0.0.4</pre> Pinging 10.0.0.4 with 32 bytes of data: <pre>Reply from 10.0.0.4: bytes=32 time=17ms TTL=254 Reply from 10.0.0.4: bytes=32 time=16ms TTL=254 Reply from 10.0.0.4: bytes=32 time=17ms TTL=254 Reply from 10.0.0.4: bytes=32 time=16ms TTL=254</pre> Ping statistics for 10.0.0.4: <pre>Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 16ms, Maximum = 18ms, Average = 16ms</pre>
Gamer_Hyderabad to ISP_Hyderabad	Gamer_Chennai to ISP_Chennai

Fig. 12. Gamer Laptops to Local ISP

<pre>C:\>ping 172.16.1.1</pre> Pinging 172.16.1.1 with 32 bytes of data: <pre>Reply from 172.16.1.1: bytes=32 time=25ms TTL=251 Reply from 172.16.1.1: bytes=32 time=17ms TTL=251 Reply from 172.16.1.1: bytes=32 time=5ms TTL=251 Reply from 172.16.1.1: bytes=32 time=18ms TTL=251</pre> Ping statistics for 172.16.1.1: <pre>Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 5ms, Maximum = 25ms, Average = 17ms</pre>	<pre>C:\>ping 172.16.1.1</pre> Pinging 172.16.1.1 with 32 bytes of data: <pre>Reply from 172.16.1.1: bytes=32 time=23ms TTL=251 Reply from 172.16.1.1: bytes=32 time=16ms TTL=251 Reply from 172.16.1.1: bytes=32 time=19ms TTL=251 Reply from 172.16.1.1: bytes=32 time=15ms TTL=251</pre> Ping statistics for 172.16.1.1: <pre>Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 16ms, Maximum = 23ms, Average = 19ms</pre>
Gamer_Bangalore to Singapore Server <pre>C:\>ping 172.16.1.1</pre> Pinging 172.16.1.1 with 32 bytes of data: <pre>Reply from 172.16.1.1: bytes=32 time=40ms TTL=251 Reply from 172.16.1.1: bytes=32 time=23ms TTL=251 Reply from 172.16.1.1: bytes=32 time=10ms TTL=251 Reply from 172.16.1.1: bytes=32 time=9ms TTL=251</pre> Ping statistics for 172.16.1.1: <pre>Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 6ms, Maximum = 23ms, Average = 13ms</pre>	Gamer_Delhi to Singapore Server <pre>C:\>ping 172.16.1.1</pre> Pinging 172.16.1.1 with 32 bytes of data: <pre>Reply from 172.16.1.1: bytes=32 time=40ms TTL=251 Reply from 172.16.1.1: bytes=32 time=10ms TTL=251 Reply from 172.16.1.1: bytes=32 time=7ms TTL=251</pre> Ping statistics for 172.16.1.1: <pre>Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 7ms, Maximum = 20ms, Average = 13ms</pre>
Gamer_Hyderabad to Singapore Server	Gamer_Chennai to Singapore Server

Fig. 13. Successful Ping to Singapore Server

2) *Packet Flow Simulation:* The simulation mode in Cisco Packet Tracer was used to simulate the movement of packets from Gamer Laptops to the Singapore Server and the path back from the Singapore Server to Gamer Laptops.

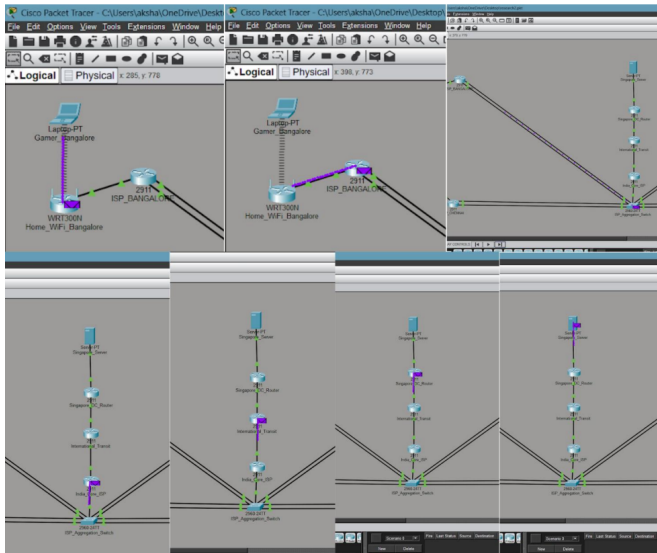


Fig. 14. Gamer Laptops to Singapore Server

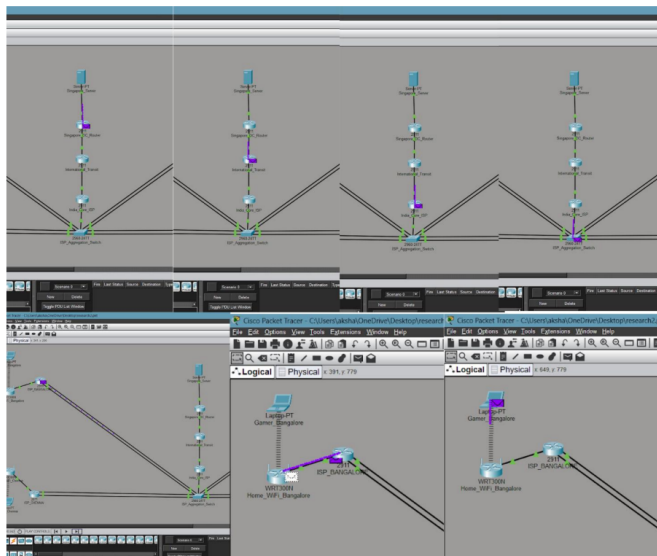


Fig. 15. Singapore Server to Gamer Laptops

B. Scenario 2 Connectivity Testing

After deployment of a Modular Edge Data Center, all the gamer laptops were directed to it via better routing protocols and was confirmed by pinging it from all the gamer laptops.

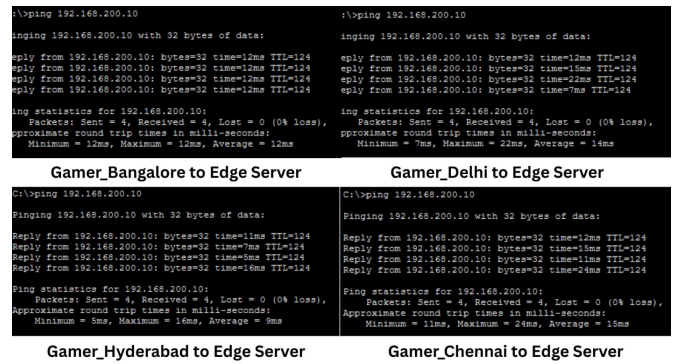


Fig. 16. Gamer Laptops to Edge Server

1) *Packet Flow Simulation:* Simulation mode in Cisco Packet Tracer was used to check and confirm the packet flow.

Traditional path (Scenario 1):

Gamer → ISP → India Core → International → Singapore

Optimized path (Scenario 2):

Gamer → ISP → India Core → MDC → Edge Server

The optimized path showed significantly less hops and improved user experience.

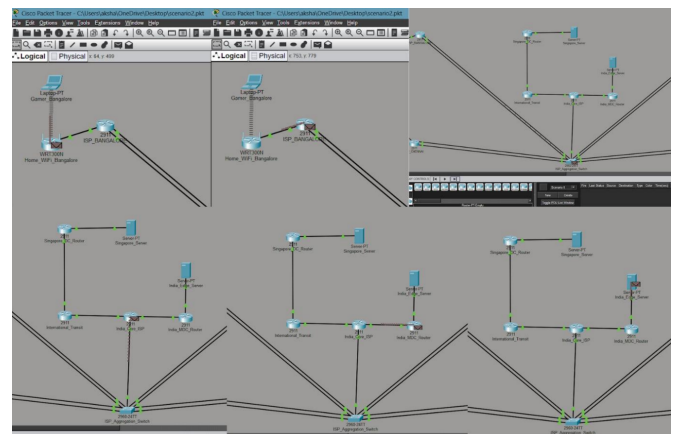


Fig. 17. Packet Flow to Edge Server

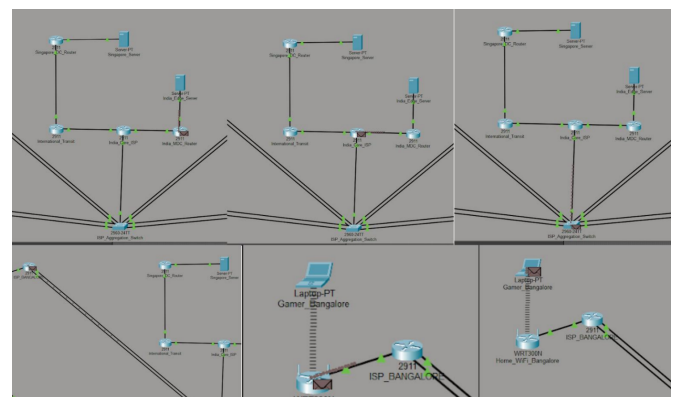


Fig. 18. Packet Flow from Edge Server

VI. DISCUSSION

A. Final Comparison

TABLE II
SCENARIO COMPARISON

Parameter	Scenario 1	Scenario 2
Server Location	Singapore	Local Edge + Cloud
Routing Distance	International	Localized
Hop Count	High	Reduced
Latency (Simulated)	~20 ms	10-15 ms
Latency (Real Expected)	High	Significantly Reduced
Performance	Moderated	Optimized
Scalability	Limited	High
User Experience	Significant Delay	Low Latency Performance

B. Limitations of Cisco Packet Tracer

While simulating both the scenarios, a common phenomenon was observed when the gamer laptops were pinging on the server. We saw a negligible or very minute difference in the latency in the scenarios which is completely opposite of what we are trying to achieve by this paper. But this issue was due to the inherent limitations of Cisco Packet Tracer. Cisco Packet Tracer is only used to simulate:

- Network topology
- Routing protocols
- Packet flow
- Logical connectivity

However, it cannot accurately simulate real-world geographical distance. Cisco Packet Tracer calculates latency on hop counts rather than real life distance. As a result, the software treats all the devices to be in the same geographical location. The physical distance between the places is not translated into reality. The need for this simulation was to demonstrate just the architectural and routing optimization. The proposed MDC architecture reduces:

- Routing distance
- Number of network hops
- Dependency on international servers
- Network congestion potential

C. Impact of Edge-Based Architecture on Latency-Sensitive Applications

The results we got from the simulation tell us the importance of the proposed architectural design in helping latency dependent applications such as online multiplayer games provide a better experience for the players. In Scenario 1, which displays the traditional architecture, the network traffic must travel through various hops and routing protocols before it can reach its destination. This leads to an increased routing distance which causes delays and degraded user experience. Therefore, in Scenario 2 as we saw the proposed architecture of Micro Data Centers, an Edge server is placed within the national network directly connected to the India Core ISP. Therefore, the network path is directed towards it except directly going to the International Transit. We saw this reduces the routing distance resulting in low latency, reduced jitter, and improved user experience.

D. Role of Optimized Routing in Traffic Engineering

Another observation we noticed in the simulation is that optimized routing improves network efficiency. The use of OSPF routing allows traffic to be directed to the nearest server. By assigning lower cost to the Edge Server and higher cost to International Transit, the protocol automatically routes to the lowest cost possible. This mechanism tells us how cost-based routing strategies can be used for intelligent traffic routing in real life network scenarios.

E. Implications for Cloud Gaming Infrastructure in India

The results we have observed in this paper are very similar to the Indian Digital Ecosystem. Although internet connectivity has penetrated the backbone of the country, many data servers remain in metropolitan cities. Therefore, for users from Tier 2 and Tier 3 cities users must connect to these distant servers mostly located very far or not even in this country as we saw in Scenario 1 where the main cloud server was in Singapore. The proposed architecture in Scenario 2 can solve this problem. By setting up Micro Data Centers within the country, all the traffic is processed within rather than cross international boundaries. This method really improves user experience and reduces network traffic congestion.

VII. CONCLUSION

This paper talked about the latency challenges associated with traditional cloud architectures and talked about a proposed architecture by integrating Modular Edge Data Centers. Using Cisco Packet Tracer, both scenarios were simulated to evaluate the impact of regional server placement and optimized routing. In the first scenario we simulated a traditional cloud gaming architecture where users living in multiple Indian cities try to connect to a centralized server located in Singapore. The Cisco Packet Tracer simulated demonstrates that this architecture needs multiple hops and routing layers to reach the desired address that increases the path length and routing complexity. The second scenario introduces an edge-based architecture by deploying a Micro Data Server connected to India Core Router so that the traffic need not use the international transit and go to Singapore server instead the MDC is placed within the national network infrastructure. By using OSPF routing, the network traffic is directed towards the edge server installed instead of international transit reducing the hop count and the routing distance. After simulating both the scenarios, the results proved that the proposed architecture in Scenario 2 reduces the routing layers and hops. Although Cisco Packet Tracer does not replicate real life geographical distance or latency, the reduced number of routing layers gives strong evidence that the edge servers can improve latency sensitive applications such as the multiplayer online games.

VIII. REFERENCES

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